

Przetwarzanie i analiza danych przestrzennych Oracle spatial

1. Połączenie z bazą Oracle

SQL developer

Parametry połączenia z bazą Oracle na serwerze II (wymagane jest dodatkowo połączenie VPN z siecią AGH)

New / Select Database Connection

Name:

Database Type:

User Info Proxy User

Authentication Type:

Username: Role:

Password: ☒ Save Password

Connection Type:

Details Advanced

Hostname:

Port:

☒ SID:

☐ Service name:

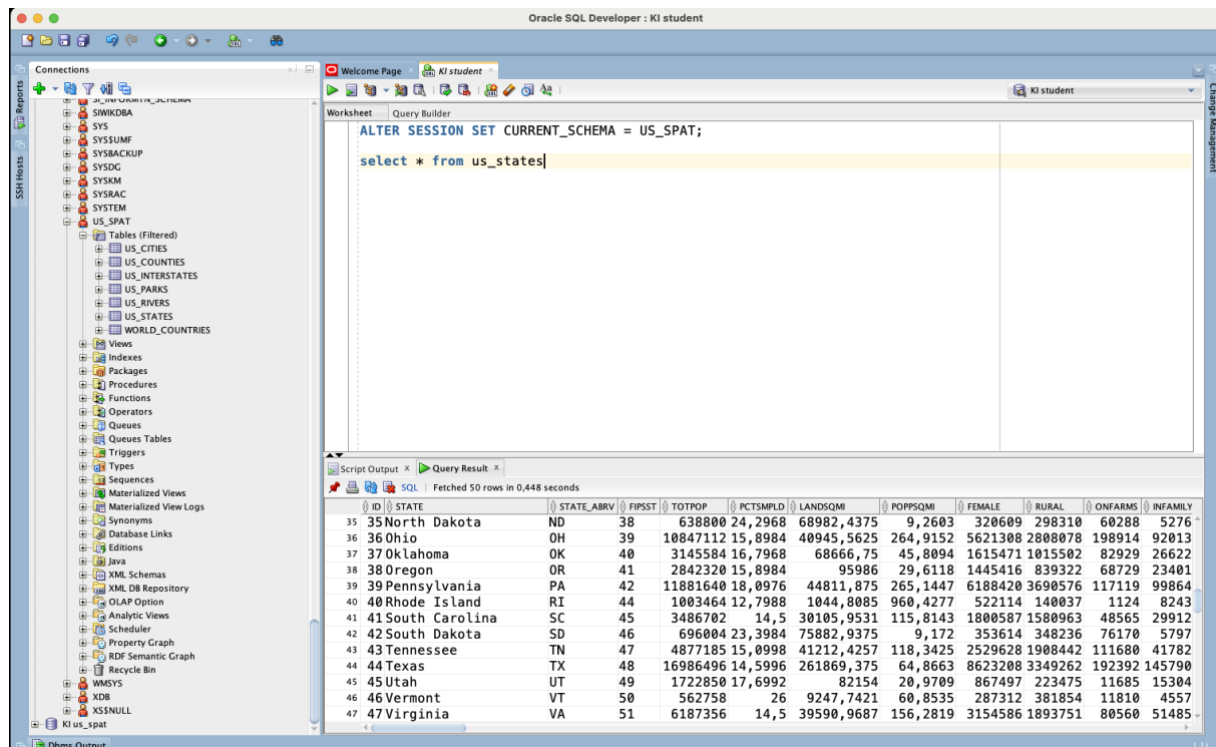
Po nawiązaniu połączenia należy wybrać schemat US_SPAT

```
ALTER SESSION SET CURRENT_SCHEMA = US_SPAT;
```

Następnie można wykonać polecenie SQL odwołujące się do jednej z przykładowych tabel

Np.

```
SELECT * FROM us_states
```



The screenshot shows the Oracle SQL Developer interface. The left pane displays the 'Connections' tree with 'US_SPAT' selected. The main window shows a SQL script with the following content:

```
ALTER SESSION SET CURRENT_SCHEMA = US_SPAT;  
select * from us_states
```

The 'Query Result' pane at the bottom displays the results of the query, showing 50 rows. The data is as follows:

ID	STATE	STATE_ABRV	FIPSST	TOTPOP	PCTSMPLD	LANDSQMI	POPPSQMI	FEMALE	RURAL	ONFARMS	INFAMILY
35	North Dakota	ND	38	638800	24,2968	68982,4375	9,2603	320609	298310	60288	5276
36	Ohio	OH	39	10847112	15,8984	40945,5625	264,9152	5621308	2808078	198914	92013
37	Oklahoma	OK	40	3145584	16,7968	68666,75	45,8094	1615471	1015502	82929	26622
38	Oregon	OR	41	2842320	15,8984	95986	29,6118	1445416	839322	68729	23401
39	Pennsylvania	PA	42	11881640	18,0976	44811,875	265,1447	6188420	3690576	117119	99864
40	Rhode Island	RI	44	1003464	12,7988	1044,8085	960,4277	522114	140037	1124	8243
41	South Carolina	SC	45	3486702	14,5	30105,9531	115,8143	1800587	1580963	48565	29912
42	South Dakota	SD	46	696004	23,3984	75882,9375	9,172	353614	348236	76170	5797
43	Tennessee	TN	47	4877185	15,0998	41212,4257	118,3425	2529628	1908442	111680	41782
44	Texas	TX	48	16986496	14,5996	261869,375	64,8663	8623208	3349262	192392	145790
45	Utah	UT	49	1722850	17,6992	82154	20,9709	867497	223475	11685	15304
46	Vermont	VT	50	562758	26	9247,7421	60,8535	287312	381854	11810	4557
47	Virginia	VA	51	6187356	14,5	39590,9687	156,2819	3154586	1893751	80560	51485

2. Przykładowa baza danych przestrzennych

Zapoznaj się ze strukturą bazy danych

Tabele:

us_cities

us_counties

us_interstates

us_parks

us_rivers

us_states

world_countries

w panelu po lewej stronie należy wybrać zakładkę other_users a następnie schemat/użytkownika us_spat

The screenshot shows the Oracle SQL Developer interface. On the left, the 'Connections' pane displays the 'us_states' table under the 'Other Users' schema. A red arrow points to this table. The main window shows the 'Query Builder' with the following SQL query:

```
ALTER SESSION SET CURRENT_SCHEMA = US_SPAT;  
select * from us_states
```

The 'Query Result' pane displays the following data:

ID	STATE	STATE_ABRV	FIPSST	TOTPOP	PCTSMPLD	LANDSQMI	POPPSQMI	FEMALE	RURAL	ONFARMS	INFAMILY
35	North Dakota	ND	38	638800	24,2968	68982,4375	9,2603	320609	298310	60288	5276
36	Ohio	OH	39	10847112	15,8984	40945,5625	264,9152	5621308	2808078	198914	92013
37	Oklahoma	OK	40	3145584	16,7968	68666,75	45,8094	1615471	1015502	82929	26622
38	Oregon	OR	41	2842320	15,8984	95986	29,6118	1445416	839322	68729	23401
39	Pennsylvania	PA	42	11881640	18,0976	44811,875	265,1447	6188420	3690576	117119	99864
40	Rhode Island	RI	44	1003464	12,7988	1044,8085	960,4277	522114	140037	1124	8243
41	South Carolina	SC	45	3486702	14,5	30105,9531	115,8143	1800587	1580963	48565	29912
42	South Dakota	SD	46	696004	23,3984	75882,9375	9,172	353614	348236	76170	5797
43	Tennessee	TN	47	4877185	15,0998	41212,4257	118,3425	2529628	1908442	111680	41782
44	Texas	TX	48	16986496	14,5996	261869,375	64,8663	8623208	3349262	192392	145790
45	Utah	UT	49	1722850	17,6992	82154	20,9709	867497	223475	11685	15304
46	Vermont	VT	50	562758	26	9247,7421	60,8535	287312	381854	11810	4557
47	Virginia	VA	51	6187356	14,5	39590,9687	156,2819	3154586	1893751	80560	51485

Oracle SQL Developer

Connections

- SSB Hosts
 - SWIKDBA
 - SYS
 - SYS\$SUMF
 - SYSBACKUP
 - SYSDBG
 - SYSKM
 - SYSRAC
 - SYSTEM
 - US_SPAT
 - Tables (Filtered)
 - US_CITIES
 - US_COUNTIES
 - US_INTERSTATES
 - US_PARKS
 - US_RIVERS
 - US_STATES
 - WORLD_COUNTRIES
 - Views
 - Indexes
 - Packages
 - Procedures
 - Functions
 - Operators
 - Queues
 - Queues Tables
 - Triggers
 - Types
 - Sequences
 - Materialized Views
 - Materialized View Logs
 - Synonyms
 - Database Links
 - Editions
 - Java
 - XML Schemas
 - XML DB Repository
 - OLAP Option
 - Analytic Views
 - Scheduler
 - Property Graph
 - RDF Semantic Graph
 - Recycle Bin
 - WMSYS
 - XDB
 - X\$NULL
 - KI us_spat

Worksheet

```
ALTER SESSION SET CURRENT_SCHEMA = US_SPAT;  
  
select * from us_states
```

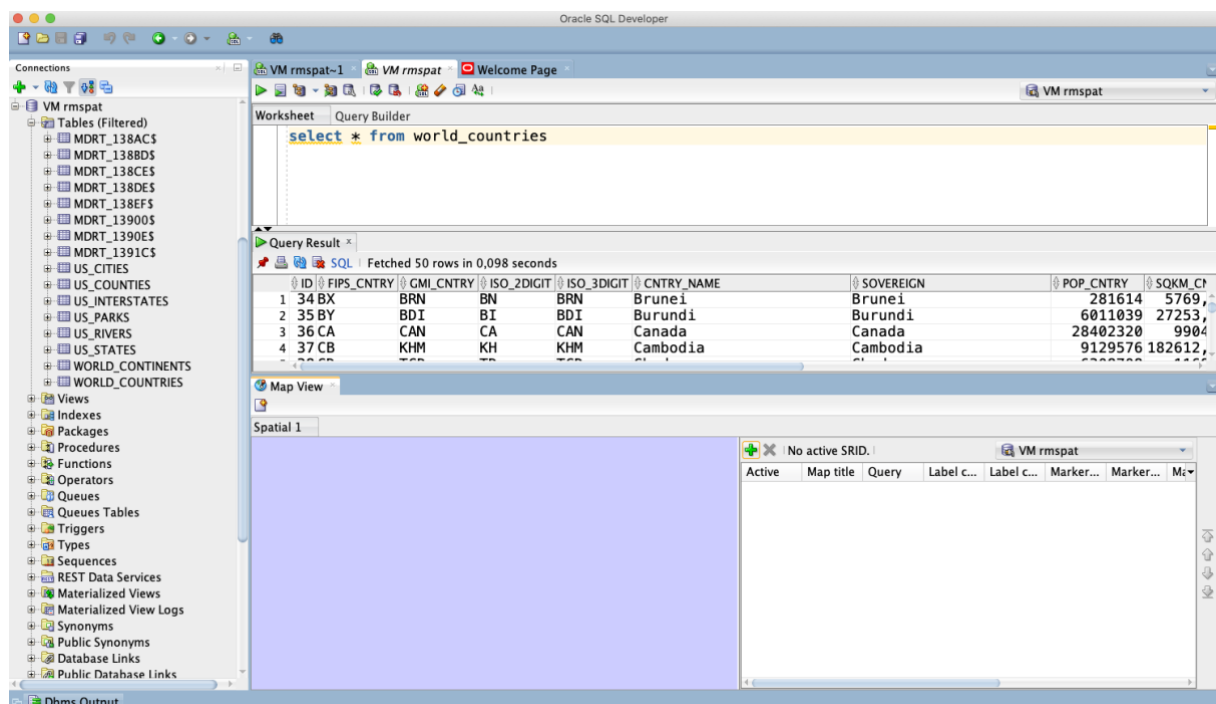
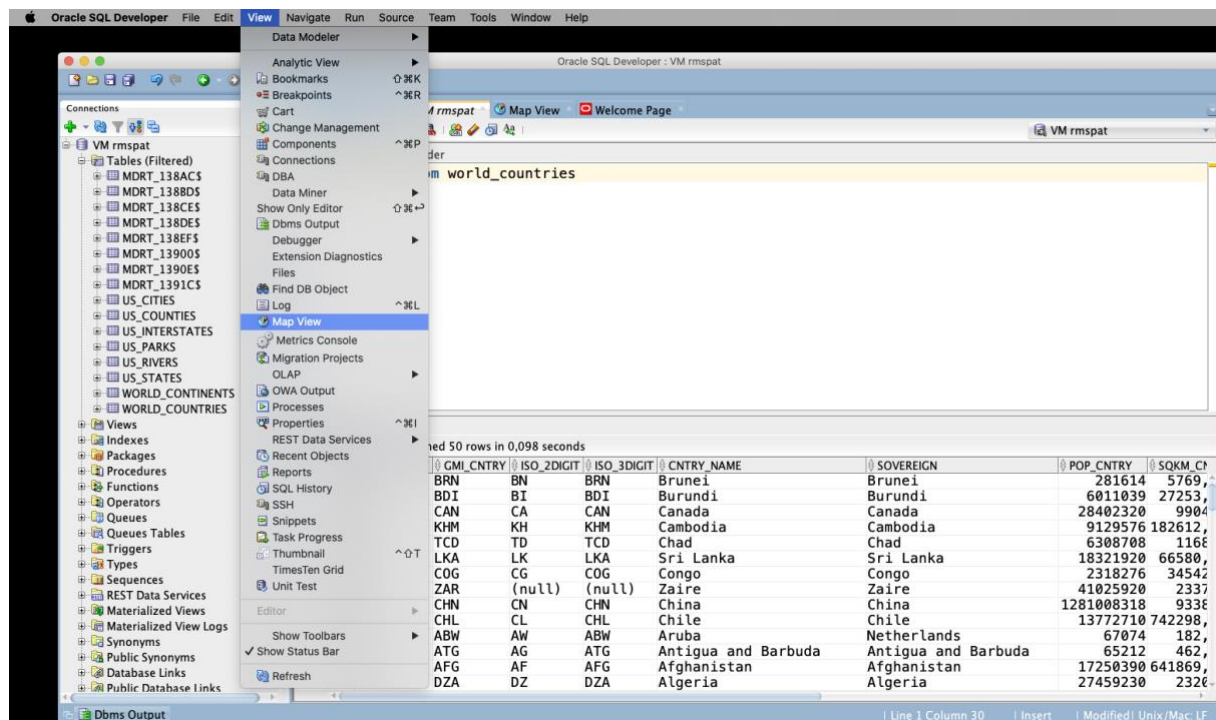
Script Output x Query Result x

SQL | Fetched 50 rows in 0,448 seconds

ID	STATE	STATE_ABRV	FIPSST	TOTPOP	PCTSMPLD	LANDSOMI	POPPSQMI	FEMALE	RURAL	ONFARMS	INFAMILY
35	North Dakota	ND	38	638800	24,2968	68982,4375	9,2603	320609	298310	60288	5276
36	Ohio	OH	39	10847112	15,8984	40945,5625	264,9152	5621308	2808078	198914	92013
37	Oklahoma	OK	40	3145584	16,7968	68666,75	45,8094	1615471	1015502	82929	26622
38	Oregon	OR	41	2842320	15,8984	95986	29,6118	1445416	839322	68729	23401
39	Pennsylvania	PA	42	11881640	18,0976	44811,875	265,1447	6188420	3690576	117119	99864
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47	Virginia	VA	51	6187356	14,5	39590,9687	156,2819	3154586	1893751	80560	51485

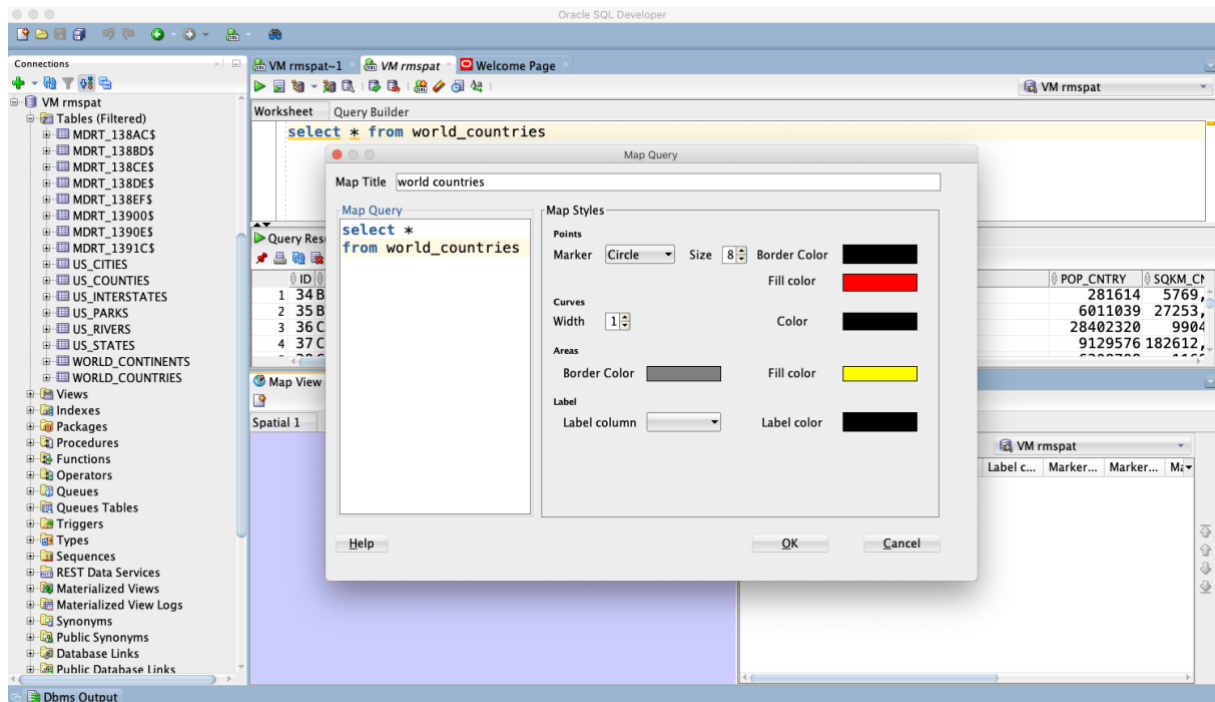
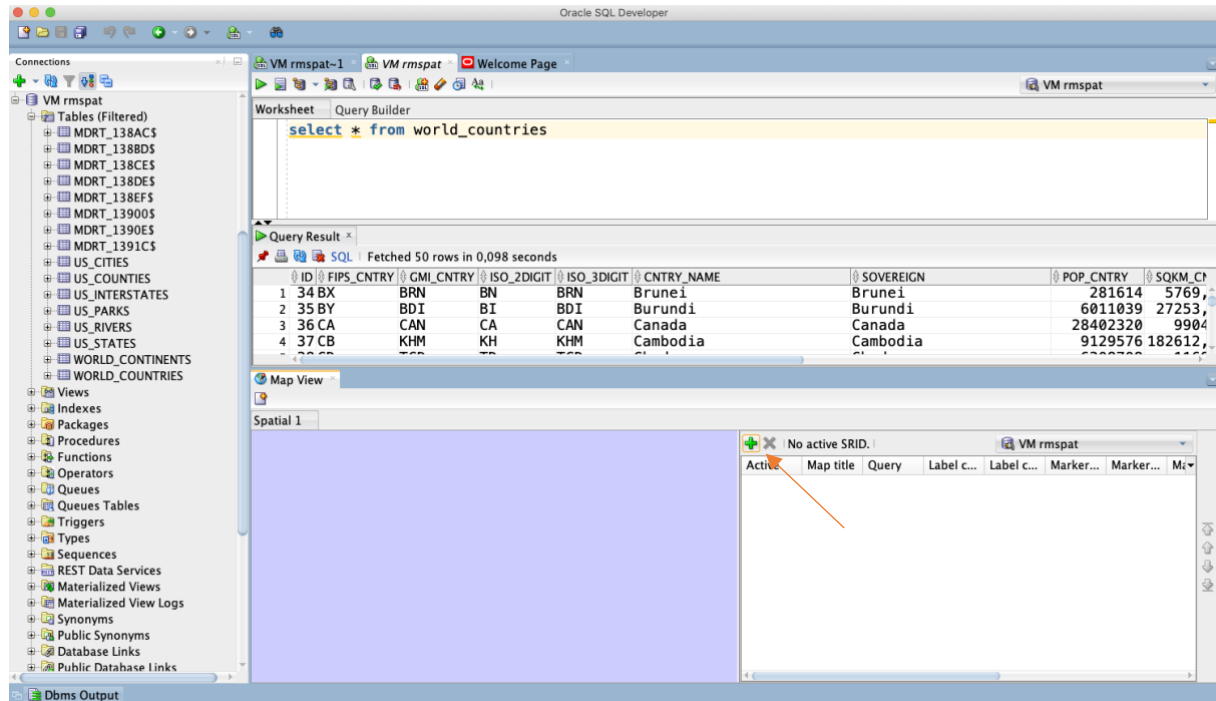
3. Map View

Uruchom widok mapy – (w menu wybieramy View/Map View)



Zwizualizuj wynik polecenia

```
SELECT *  
FROM world_countries
```



Oracle SQL Developer

Connections: VM rmspat-1, VM rmspat, Welcome Page

VM rmspat

Tables (Filtered): MDRT_138AC\$, MDRT_138BD\$, MDRT_138CE\$, MDRT_138DE\$, MDRT_138EF\$, MDRT_13900\$, MDRT_1390E\$, MDRT_1391C\$, US_CITIES, US_COUNTIES, US_INTERSTATES, US_PARKS, US_RIVERS, US_STATES, WORLD_CONTINENTS, WORLD_COUNTRIES

Views, Indexes, Packages, Procedures, Functions, Operators, Queues, Queues Tables, Triggers, Types, Sequences, REST Data Services, Materialized Views, Materialized View Logs, Synonyms, Public Synonyms, Database Links, Public Database Links

Dbms Output

Worksheet: Query Builder

```
select * from world_countries
```

Query Result: Fetched 50 rows in 0,098 seconds

ID	FIPS_CNTRY	GMI_CNTRY	ISO_2DIGIT	ISO_3DIGIT	CNTRY_NAME	SOVEREIGN	POP_CNTRY	SQKM_CN
1	34 BX	BRN	BN	BRN	Brunei	Brunei	281614	5769
2	35 BY	BDI	BI	BDI	Burundi	Burundi	6011039	27253
3	36 CA	CAN	CA	CAN	Canada	Canada	28402320	9904
4	37 CB	KHM	KH	KHM	Cambodia	Cambodia	9129576	182612

Map View: Spatial 1

Active SRID: 8307

Map title: world co... select * ...

Query: world co... select * ...

Label c...:

Label c...:

Marker...: Circle

Marker...:

Ma...:

Inny sposób

The screenshot shows the Oracle SQL Developer interface. The 'Connections' pane on the left lists various database objects under the 'VM rmspat' connection. The 'Worksheet' pane displays the query: `select * from world_countries`. The 'Query Results' pane shows the first 14 rows of the query result. A context menu is open over the row for 'Sri Lanka' (row 6), with the option 'Invoke Map View on result set' highlighted.

ID	FIPS_CNTRY	GMI_CNTRY	ISO_2DIGIT	ISO_3DIGIT	CNTRY_NAME	SOVEREIGN	POP_CNTRY	SQKM_CNTRY
1	34 BX	BRN	BN	BRN	Brunei	Brunei	281614	5769
2	35 BY	BDI	BI	BDI	Burundi	Burundi	6011039	27253
3	36 CA	CAN	CA	CAN	Canada	Canada	28402320	9904
4	37 CB	KHM	KH	KHM	Cambodia	Cambodia	9129576	182612
5	38 CD	TCD	TD	TCD	Chad	Chad	6308708	1168
6	39 CE	LKA	LK	LKA	Sri Lanka	Sri Lanka	18321920	66580
7	40 CF	COG	CG	COG	Congo	Congo	2318276	34542
8	41 CG	ZAR	(null)	(null)	Zaire	Zaire	41025920	2337
9	42 CH	CHN	CN	CHN	China	China	1281008318	9338
10	43 CI	CHL	CL	CHL	Chile	Chile	13772710	742298
11	1 AA	ABW	AW	ABW	Aruba	Aruba	67074	182
12	2 AC	ATG	AG	ATG	Antigua and Barbuda	Antigua and Barbuda	65212	462
13	3 AF	AFG	AF	AFG	Afghanistan	Afghanistan	17250390	641869
14	4 AG	DZA	DZ	DZA	Algeria	Algeria	27459230	2320

The screenshot shows the Oracle SQL Developer interface with the 'Map View' tab selected. The 'Spatial 1' pane displays a world map with the countries highlighted in yellow. The 'Active SRID: 8307' is indicated. The 'Map title' pane shows the query: `world co... select * f...` and the 'GEOMET...' query: `select * f...`. The 'Label c...' pane shows the label: 'Circle'. The 'Marker...' pane shows the marker: 'Circle'.

Jeśli chcemy zobaczyć dane pojedynczego wiersza to:

The screenshot shows the Oracle SQL Developer interface. The 'Query Result' window displays the following data:

ID	FIPS_CNTRY	GMI_CNTRY	ISO_2DIGIT	ISO_3DIGIT	CNTRY_NAME	SOVEREIGN	POP_CNTRY
177	174 PA	PRY	PY	PRY	Paraguay	Paraguay	477346
178	176 PE	PER	PE	PER	Peru	Peru	2449640
179	190 RP	PHL	PH	PHL	Philippines	Philippines	6598112
180	175 PC	PCN	PN	PCN	Pitcairn Islands	United Kingdom	5
181	180 PL	POL	PL	POL	Poland	Poland	3791187
182	182 PO	PRT	PT	PRT	Portugal	Portugal	962551
183	191 RQ	PRI	PR	PRI	United States	United States	364793
184	186 QA	QAT	QA	QAT	Qatar	Qatar	47800
185	187 RE	REU	RE	REU	France	France	64400
186	189 RO	ROM	RO	ROM	Romania	Romania	2354055
187	192 RS	RUS	RU	RUS	Russia	Russia	15182760
188	193 RW	RWA	RW	RWA	Rwanda	Rwanda	793439
189	203 SM	SMR	SM	SMR	San Marino	San Marino	2375
190	223 TP	STP	ST	STP	Sao Tome and Principe	Sao Tome and Principe	12860

A context menu is open over the row for Poland (ID 181), showing options: 'Save Grid as Report...', 'Publish to REST', 'Invoke Map View on result set', 'Single Record View...', 'Count Rows...', 'Find/Highlight...', and 'Export...'. The 'Single Record View...' option is highlighted.

The screenshot shows the 'Single Record View' dialog box open over the 'Query Result' window. The dialog displays the following data for the selected row (ID 180):

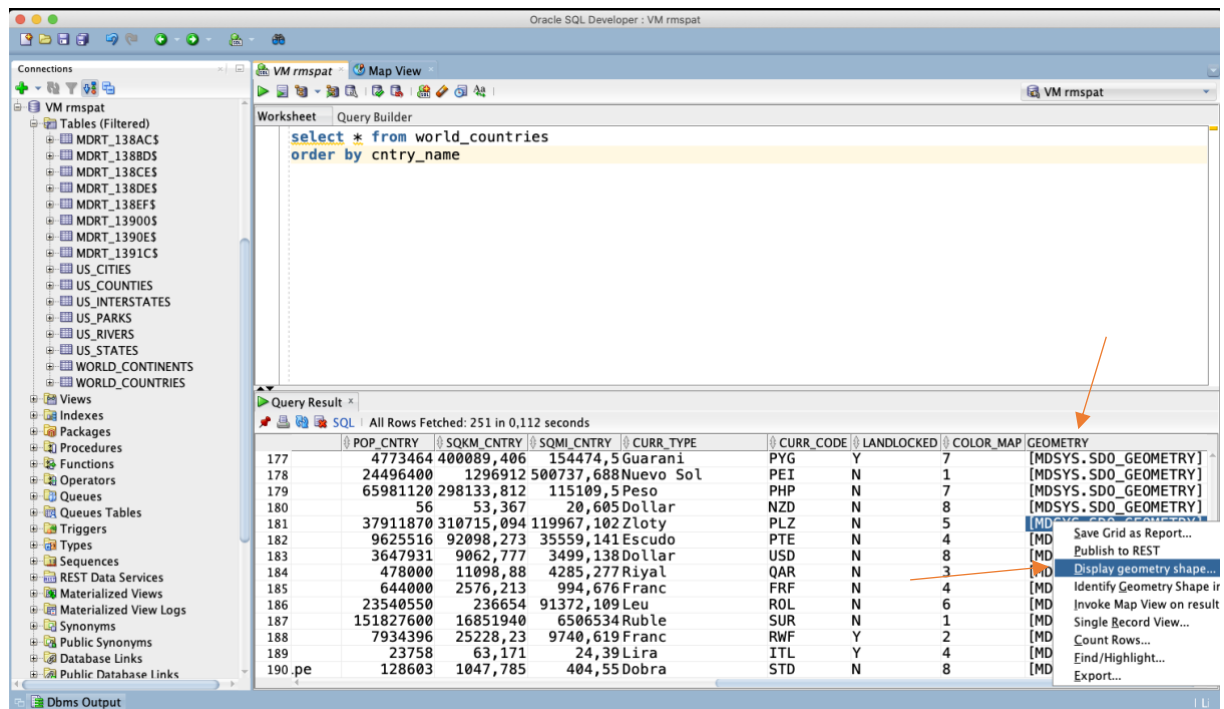
ID	FIPS_CNTRY	GMI_CNTRY	ISO_2DIGIT	ISO_3DIGIT	CNTRY_NAME	SOVEREIGN	POP_CNTRY
180	PL	POL	PL	POL	Poland	Poland	3791187

The dialog also shows additional details for the selected row:

SQKM_CNTRY	SQMI_CNTRY	CURR_TYPE	CURR_CODE
310715,094	119967,102	Zloty	PLZ

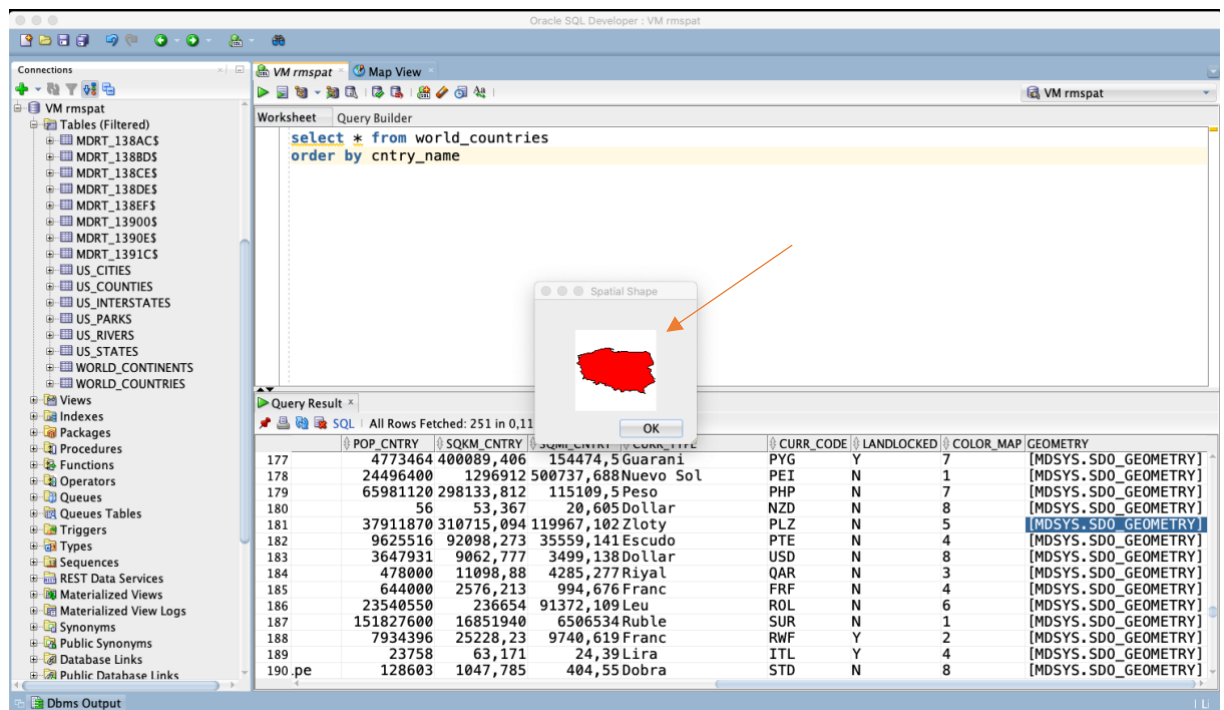
The dialog has 'Pomoć' and 'Anuluj' buttons at the bottom.

Jeśli ustawimy się na polu które zawiera geometrię to możemy ją zwizualizować



The screenshot shows the Oracle SQL Developer interface. The 'Connections' pane on the left lists various database objects. The 'Worksheet' pane contains the SQL query: `select * from world_countries order by cntry_name`. The 'Query Result' pane displays a table with columns: POP_CNTRY, SQKM_CNTRY, SQMI_CNTRY, CURR_TYPE, CURR_CODE, LANDLOCKED, COLOR_MAP, and GEOMETRY. The 'GEOMETRY' column contains values like [MDSYS.SDO_GEOMETRY]. A right-click context menu is open over the 'GEOMETRY' column, with the option 'Display geometry shape...' selected. An orange arrow points to this menu option.

	POP_CNTRY	SQKM_CNTRY	SQMI_CNTRY	CURR_TYPE	CURR_CODE	LANDLOCKED	COLOR_MAP	GEOMETRY
177	4773464	400089,406	154474,5	Guarani	PYG	Y	7	[MDSYS.SDO_GEOMETRY]
178	24496400	1296912	500737,688	Nuevo Sol	PEI	N	1	[MDSYS.SDO_GEOMETRY]
179	65981120	298133,812	115109,5	Peso	PHP	N	7	[MDSYS.SDO_GEOMETRY]
180	56	53,367	20,605	Dollar	NZD	N	8	[MDSYS.SDO_GEOMETRY]
181	37911870	310715,094	119967,102	Zloty	PLZ	N	5	[MDSYS.SDO_GEOMETRY]
182	9625516	92098,273	35559,141	Escudo	PTE	N	4	[MDSYS.SDO_GEOMETRY]
183	3647931	9062,777	3499,138	Dollar	USD	N	8	[MDSYS.SDO_GEOMETRY]
184	478000	11098,88	4285,277	Riyal	QAR	N	3	[MDSYS.SDO_GEOMETRY]
185	644000	2576,213	994,676	Franc	FRF	N	4	[MDSYS.SDO_GEOMETRY]
186	23540550	236654	91372,109	Leu	ROL	N	6	[MDSYS.SDO_GEOMETRY]
187	151827600	16851940	6506534	Ruble	SUR	N	1	[MDSYS.SDO_GEOMETRY]
188	7934396	25228,23	9740,619	Franc	RWF	Y	2	[MDSYS.SDO_GEOMETRY]
189	23758	63,171	24,39	Lira	ITL	Y	4	[MDSYS.SDO_GEOMETRY]
190	pe	128603	1047,785	Dobra	STD	N	8	[MDSYS.SDO_GEOMETRY]



The screenshot shows the same Oracle SQL Developer interface as the first image. The 'Query Result' pane displays the same table. A 'Spatial Shape' dialog box is open, showing a red polygon representing the geometry shape of the selected row. An orange arrow points to the 'Spatial Shape' dialog box.

	POP_CNTRY	SQKM_CNTRY	SQMI_CNTRY	CURR_TYPE	CURR_CODE	LANDLOCKED	COLOR_MAP	GEOMETRY
177	4773464	400089,406	154474,5	Guarani	PYG	Y	7	[MDSYS.SDO_GEOMETRY]
178	24496400	1296912	500737,688	Nuevo Sol	PEI	N	1	[MDSYS.SDO_GEOMETRY]
179	65981120	298133,812	115109,5	Peso	PHP	N	7	[MDSYS.SDO_GEOMETRY]
180	56	53,367	20,605	Dollar	NZD	N	8	[MDSYS.SDO_GEOMETRY]
181	37911870	310715,094	119967,102	Zloty	PLZ	N	5	[MDSYS.SDO_GEOMETRY]
182	9625516	92098,273	35559,141	Escudo	PTE	N	4	[MDSYS.SDO_GEOMETRY]
183	3647931	9062,777	3499,138	Dollar	USD	N	8	[MDSYS.SDO_GEOMETRY]
184	478000	11098,88	4285,277	Riyal	QAR	N	3	[MDSYS.SDO_GEOMETRY]
185	644000	2576,213	994,676	Franc	FRF	N	4	[MDSYS.SDO_GEOMETRY]
186	23540550	236654	91372,109	Leu	ROL	N	6	[MDSYS.SDO_GEOMETRY]
187	151827600	16851940	6506534	Ruble	SUR	N	1	[MDSYS.SDO_GEOMETRY]
188	7934396	25228,23	9740,619	Franc	RWF	Y	2	[MDSYS.SDO_GEOMETRY]
189	23758	63,171	24,39	Lira	ITL	Y	4	[MDSYS.SDO_GEOMETRY]
190	pe	128603	1047,785	Dobra	STD	N	8	[MDSYS.SDO_GEOMETRY]

4. Python, Jupyter notebook

Przykładowe skrypty ułatwiające rozpoczęcie pracy i analizę danych z wykorzystaniem środowiska Python, Jupyter notebook dostępne są na platformie moodle

- 1-oracle.py
 - Skrypt umożliwiający przetestowanie połączenia z bazą Oracle
- 2-oracle-spatial.py
 - Skrypt umożliwiający przetestowanie prostej analizy przestrzennej i wizualizację wyników na mapie
- 1-oracle.ipynb
 - Notebook umożliwiający przetestowanie połączenia z bazą Oracle
- 2-oracle-spatial.ipynb
 - Notebook umożliwiający przetestowanie prostej analizy przestrzennej i wizualizację wyników na mapie

Parki wewnątrz stanu Texas

```
m = folium.Map()

q = """SELECT sdo_util.to_wktgeometry(p.geom)
FROM us_parks p, us_states s
WHERE s.state = 'Texas'
AND SDO_INSIDE(p.geom, s.geom) = 'TRUE'"""

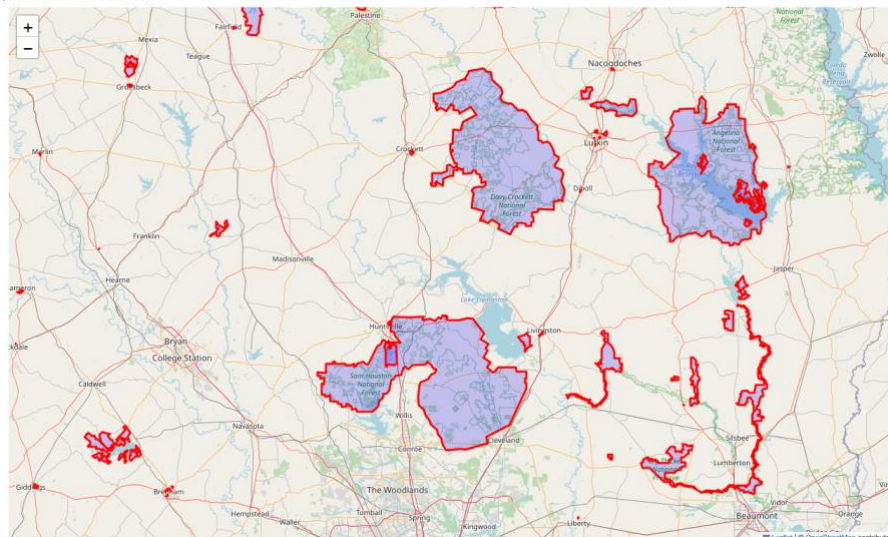
r = loads(cursor.execute(q).fetchall())

st = {'fillColor': 'blue', 'color': 'red'}

l = []
for row in r:
    g = geojson.Feature(geometry=row[0], properties={})
    l.append(g)

feature_collection = geojson.FeatureCollection(l)
folium.GeoJson(feature_collection, style_function=lambda x:st).add_to(m)
```

Executed at 2024.05.26 11:23:00 in 816ms



5. Zadania - wykonaj przykładowe zapytania - analizy przestrzenne

Dokumentacja Oracle spatial

Oracle 26

- <https://docs.oracle.com/en/database/oracle/oracle-database/26/spatl/index.html>

Oracle 19

- <https://docs.oracle.com/en/database/oracle/oracle-database/19/spatl/index.html>
-

Oracle 11/12

- <https://docs.oracle.com/database/121/SPATL/toc.htm>
- https://docs.oracle.com/cd/E18283_01/appdev.112/e11830.pdf
- <https://docs.oracle.com/database/121/SPATL/E49172-07.pdf>

Do wykonania ćwiczenia (zadania 1 – 6) i wizualizacji danych wykorzystaj Oracle SQL Developer. Alternatywnie możesz wykonać analizy w środowisku Python/Jupyter Notebook

Do wykonania zadania 7 wykorzystaj środowisko Python/Jupyter Notebook

Raport należy przesłać w formacie pdf.

Należy też dołączyć raport zawierający kod w formacie źródłowym.

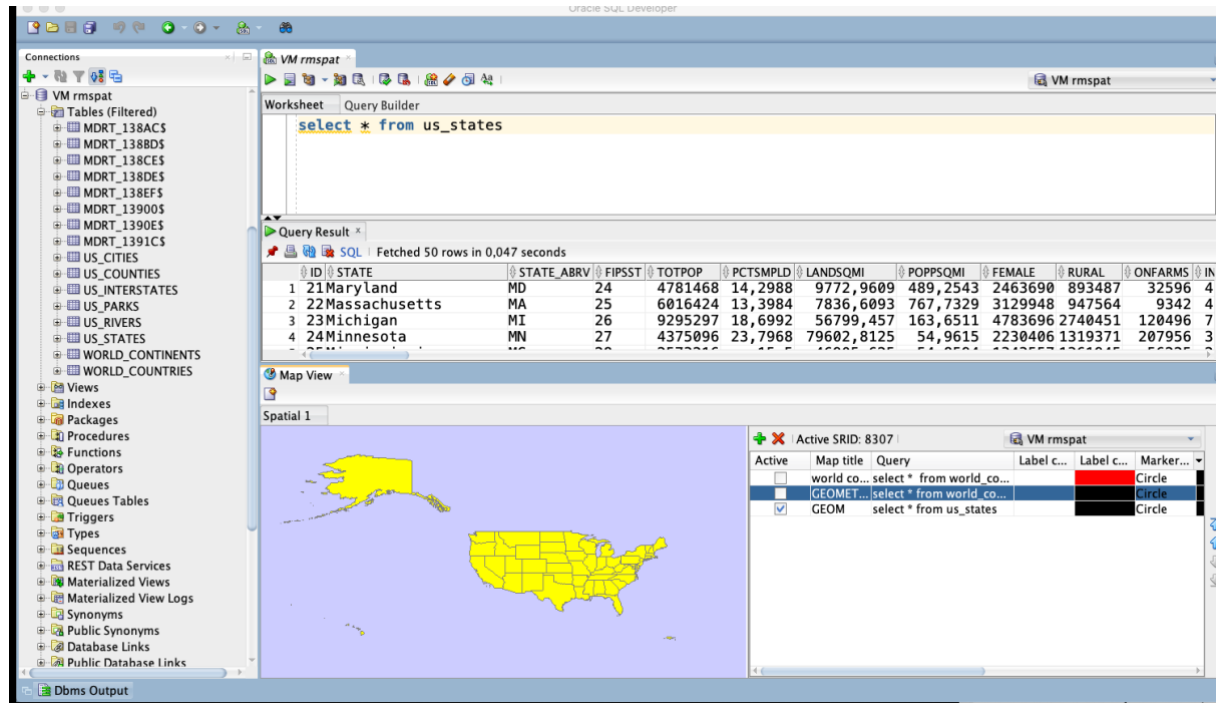
Np.

- plik tekstowy .sql z kodem poleceń
- plik .md zawierający kod wersji tekstowej
- notebook programu jupyter – plik .ipynb

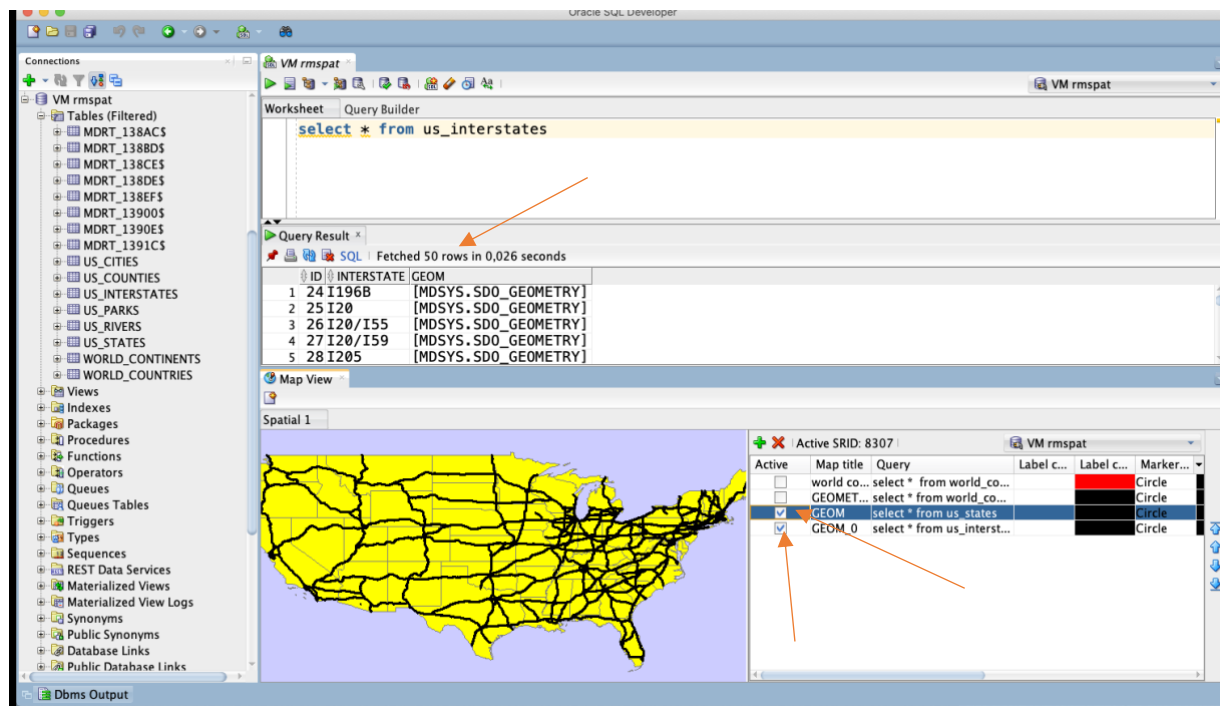
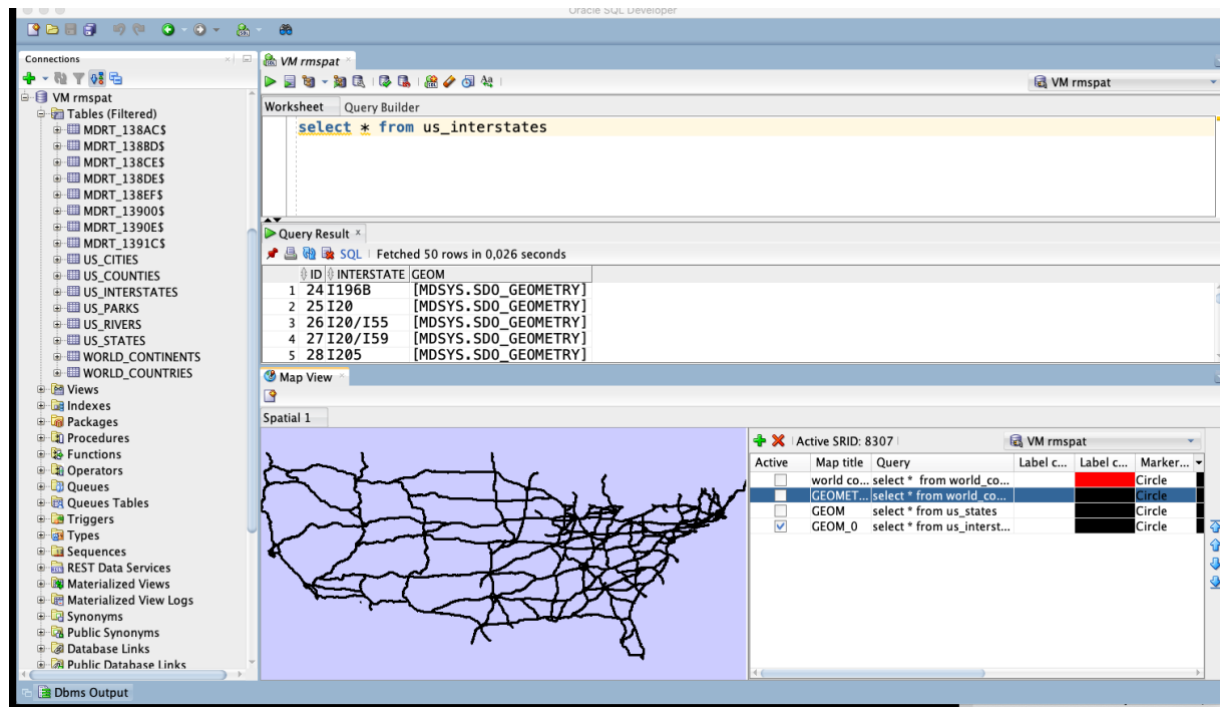
Zadanie 1

Zwizualizuj przykładowe dane

US_STATES



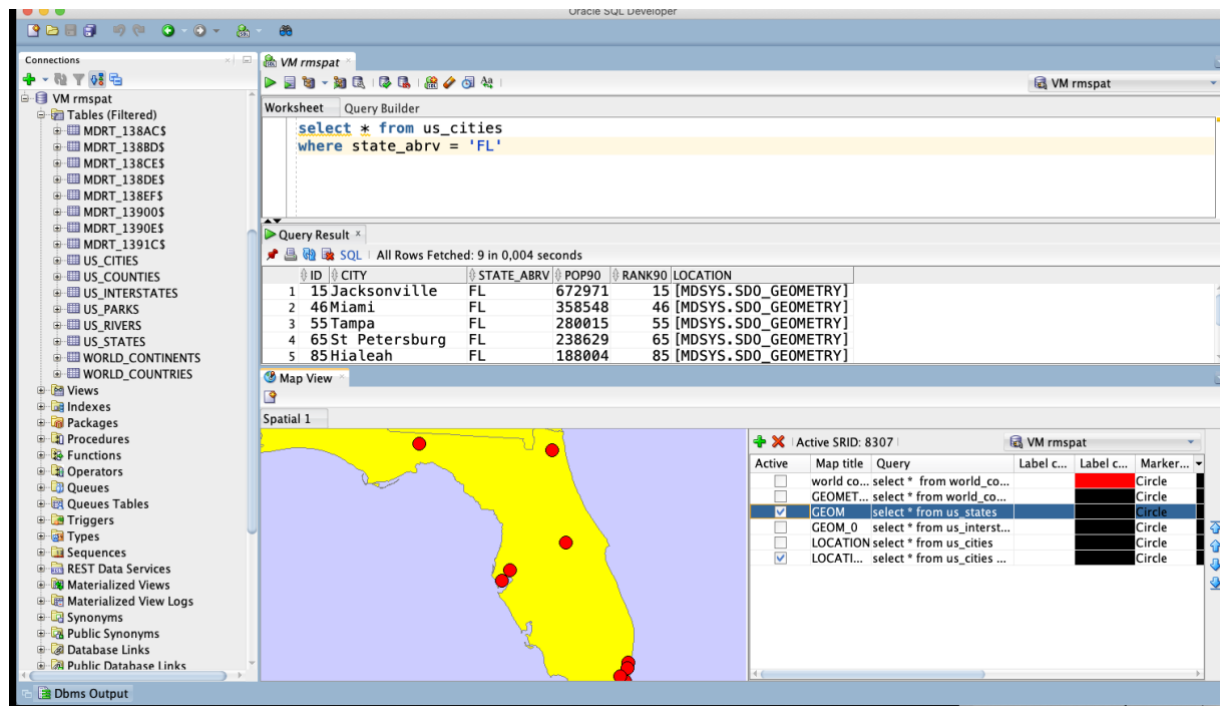
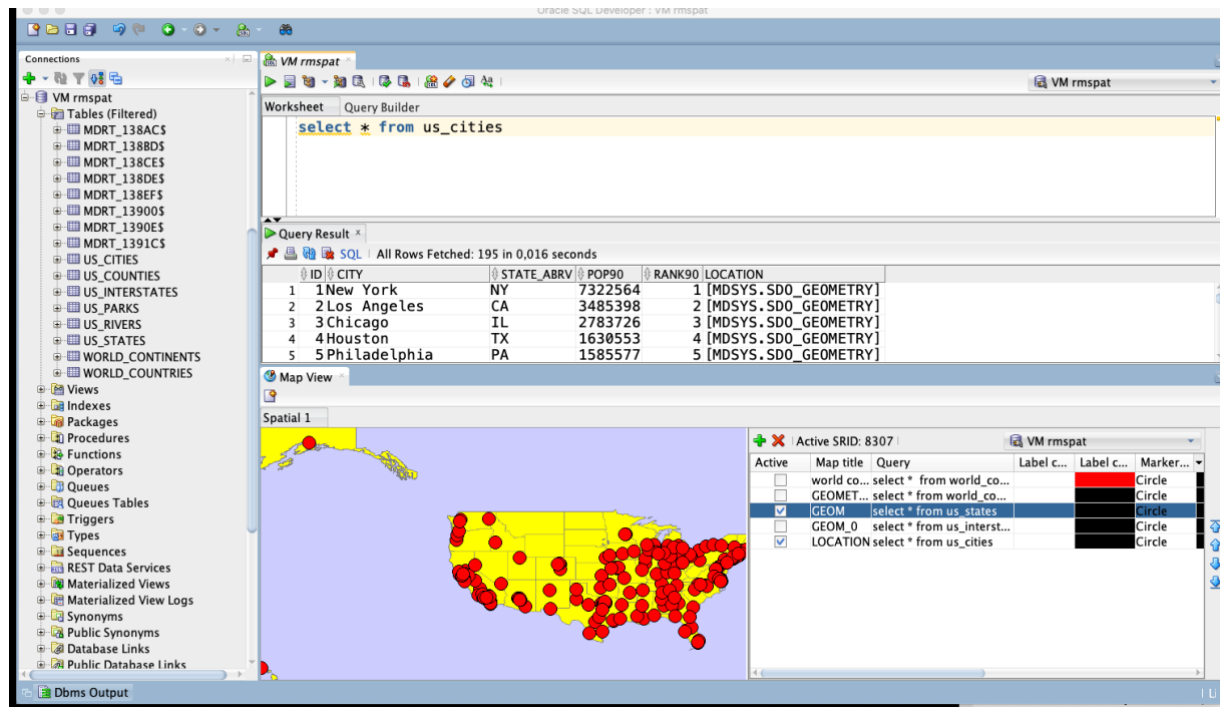
US_INTERSTATES



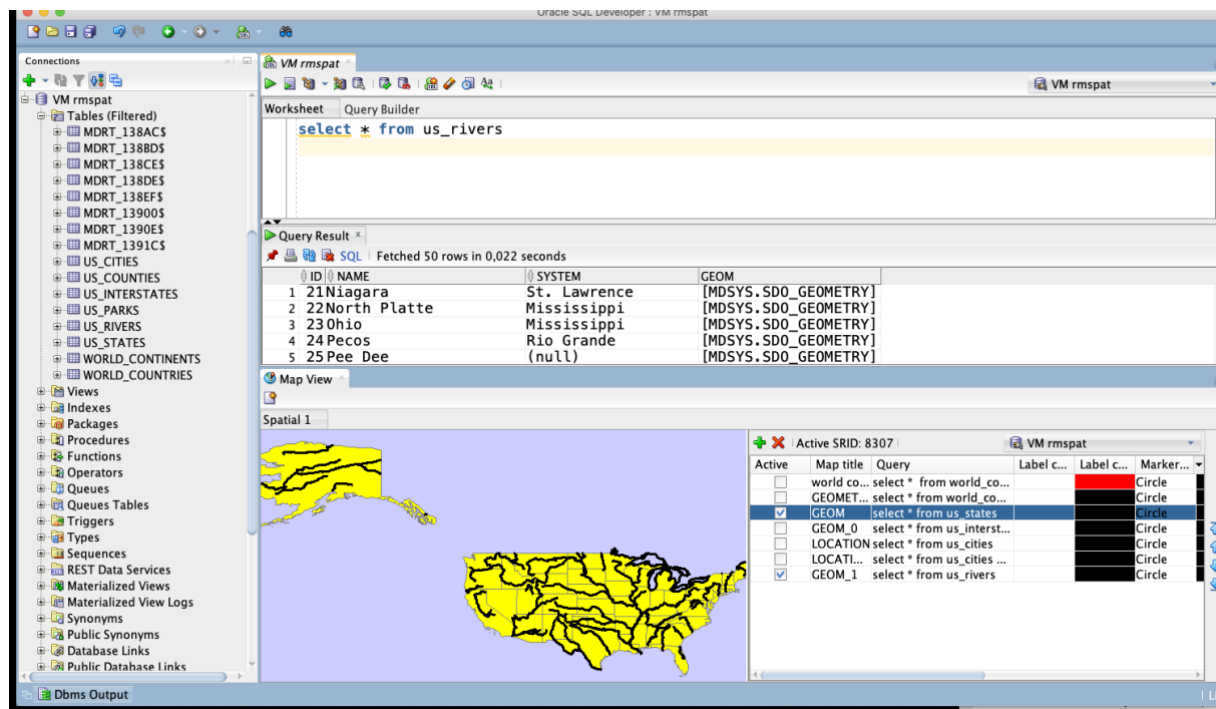
Można pokazać kilka wyników (warstw) na jednej mapie

Należy zwrócić uwagę na rozmiar zbioru danych, Jeśli zbiór danych jest zbyt duży to wizualizacja danych przy pomocy Map View może być niemożliwa.

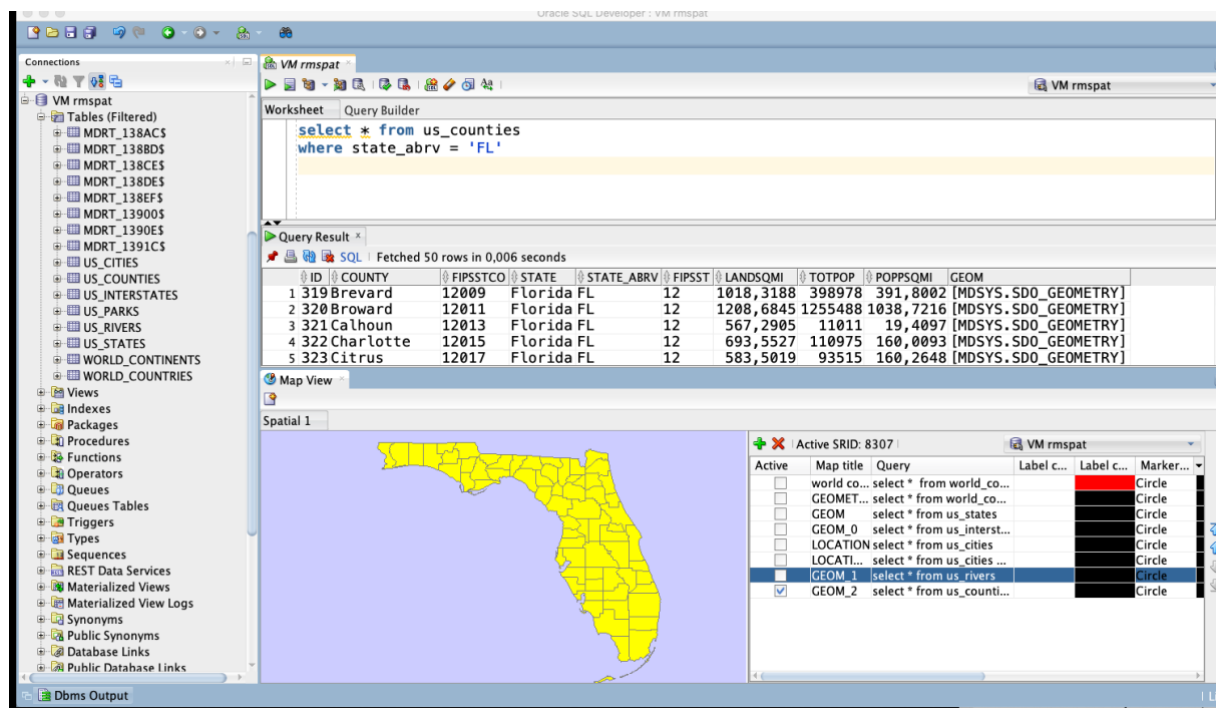
US_CITIES



US_RIVERS



US_COUNTIES



US_PARKS

The screenshot shows the QGIS desktop environment. On the left is the 'Connections' panel with a tree view of database objects. The 'VM rmspat' connection is selected, showing a list of tables including 'US_PARKS'. The main window is divided into three panes. The top pane is the 'Query Builder' showing a SQL query: `select * from us_parks where id < 50`. The middle pane is the 'Query Result' showing a table with 5 rows of data. The bottom pane is the 'Map View' showing a map of the United States with yellow markers indicating the locations of the parks. A legend on the right side of the map view shows the active SRID (8307) and a list of layers with their respective queries and markers.

Query Result:

ID	NAME	FCC	GEOM
1	Lott Park	D85	[MDSYS.SDO_GEOMETRY]
2	Joe Patrick Park	D85	[MDSYS.SDO_GEOMETRY]
3	Bluff Park	D85	[MDSYS.SDO_GEOMETRY]
4	Municipal Park	D85	[MDSYS.SDO_GEOMETRY]
5	Knohl Park	D85	[MDSYS.SDO_GEOMETRY]

Map View:

Active SRID: 8307

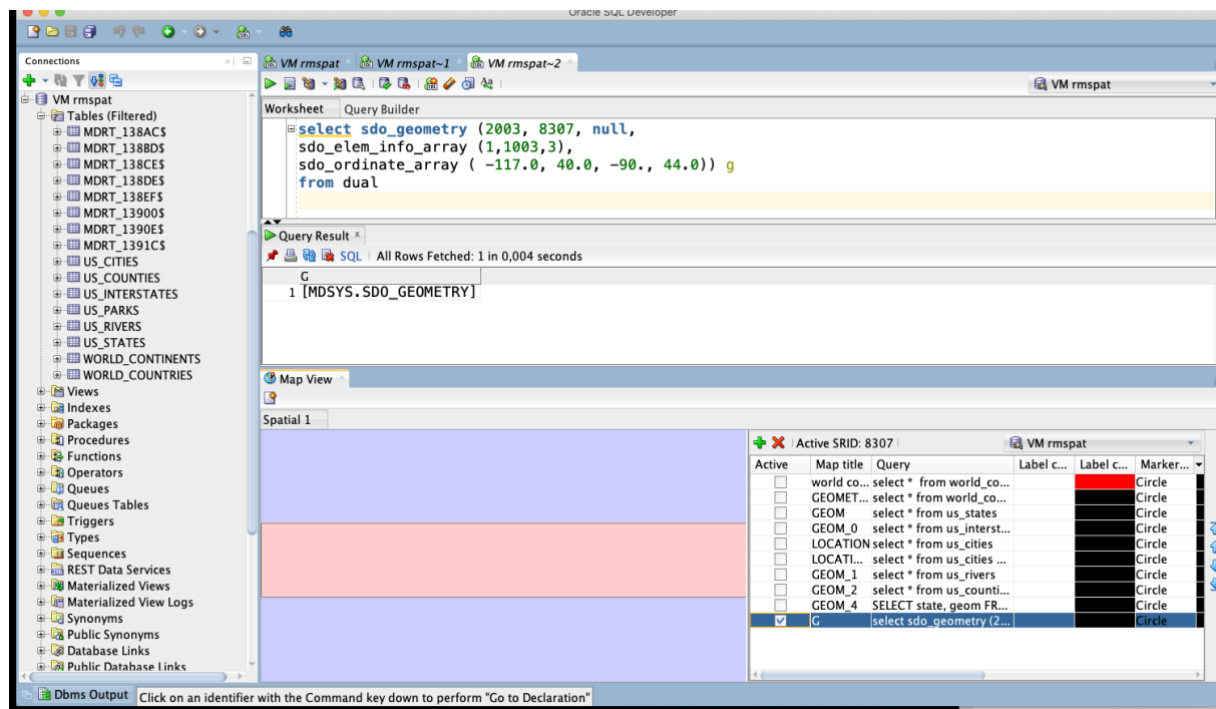
Active	Map title	Query	Label c...	Label c...	Marker...
☐	world co...	select * from world_co...			Circle
☐	GEOMET...	select * from world_co...			Circle
☒	GEOM	select * from us_parks			Circle
☐	GEOM_0	select * from us_interst...			Circle
☐	LOCATI...	select * from us_cities ...			Circle
☐	GEOM_1	select * from us_rivers ...			Circle
☐	GEOM_2	select * from us_counti...			Circle
☒	GEOM_3	select * from us_parks ...			Circle

Zadanie 2

Znajdź wszystkie stany (us_states) których obszary mają część wspólną ze wskazaną geometrią (prostokątem)

prostokąt

```
SELECT sdo_geometry (2003, 8307, null,  
sdo_elem_info_array (1,1003,3),  
sdo_ordinate_array ( -117.0, 40.0, -90., 44.0)) g  
FROM dual
```



Użyj funkcji SDO_FILTER

```
SELECT state, geom FROM us_states
WHERE sdo_filter (geom,
sdo_geometry (2003, 8307, null,
sdo_elem_info_array (1,1003,3),
sdo_ordinate_array ( -117.0, 40.0, -90., 44.0))
) = 'TRUE';
```

Zwróć uwagę na liczbę zwróconych wierszy (16)

The screenshot shows the Oracle SQL Developer interface. The 'Connections' pane on the left lists various database objects under the 'VM rmspat' connection. The 'Query Builder' pane in the center contains the following SQL query:

```
SELECT state, geom FROM us_states
WHERE sdo_filter (geom,
sdo_geometry (2003, 8307, null,
sdo_elem_info_array (1,1003,3),
sdo_ordinate_array ( -117.0, 40.0, -90., 44.0))
) = 'TRUE';
```

The 'Query Result' pane below the query shows the results of the query:

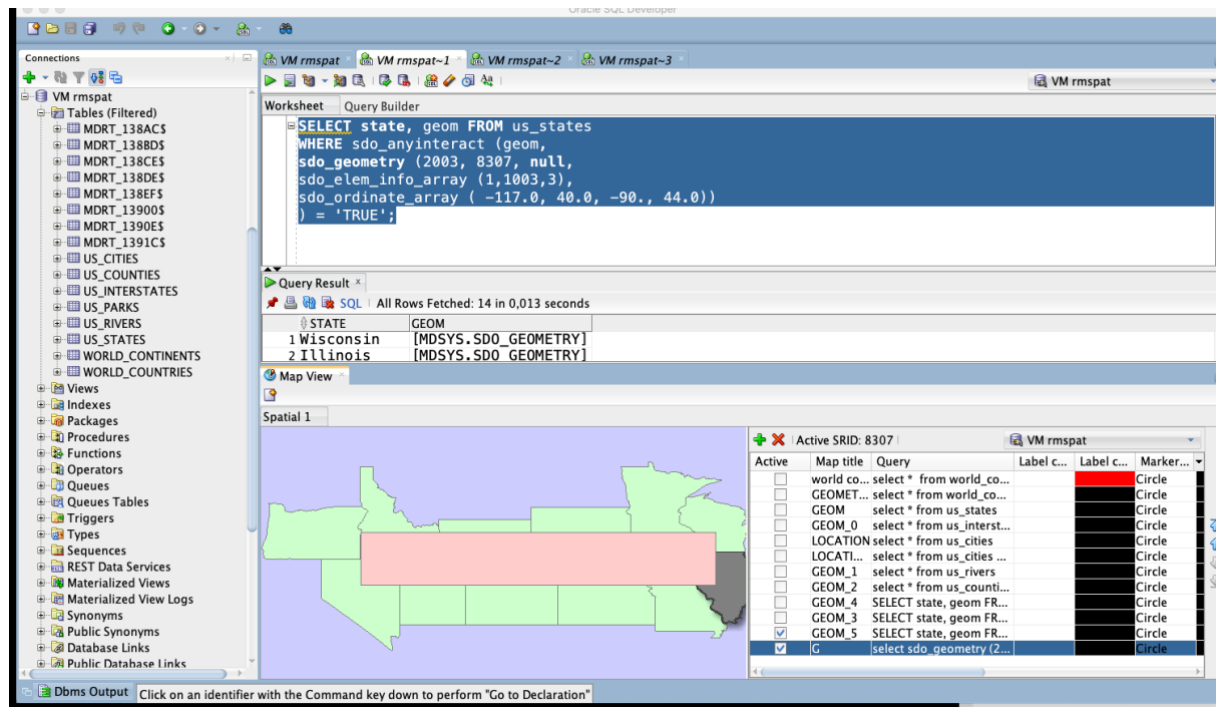
STATE	GEOM
1 Wisconsin	[MDSYS.SDO_GEOMETRY]
2 Illinois	[MDSYS.SDO_GEOMETRY]

The 'Map View' pane at the bottom shows a map of the United States with a red rectangle highlighting the area of interest. The 'Spatial 1' pane on the right shows the spatial data for the highlighted area, including a table with columns: Marker, Curve, Area Li, and Area Fill.

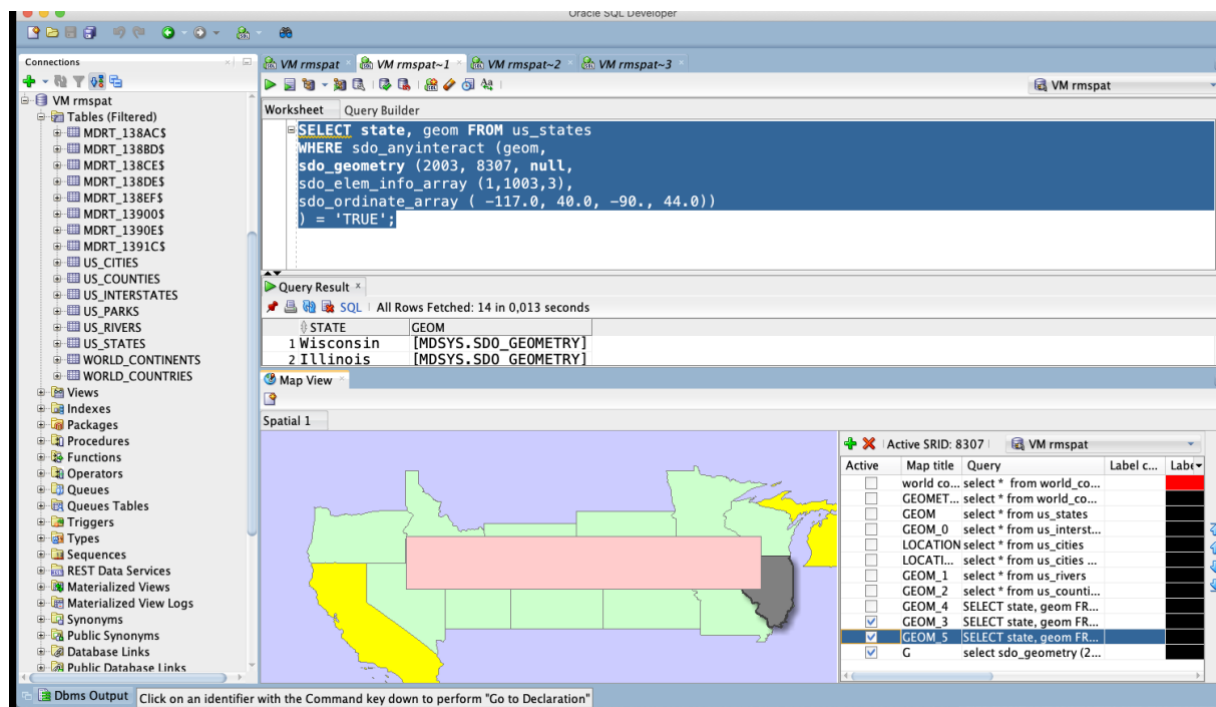
Marker	Curve	Area Li	Area Fill
8	1		
8	3		
8	3		
8	3		
8	3		
8	3		
8	3		
8	3		
8	3		
8	3		
8	3		
8	3		
8	3		
8	3		
8	3		

Użyj funkcji SDO_ANYINTERACT

```
SELECT state, geom FROM us_states
WHERE sdo_anyinteract (geom,
sdo_geometry (2003, 8307, null,
sdo_elem_info_array (1,1003,3),
sdo_ordinate_array ( -117.0, 40.0, -90., 44.0))
) = 'TRUE';
```



Porównaj wyniki sdo_filter i sdo_anyinteract



Zadanie 3

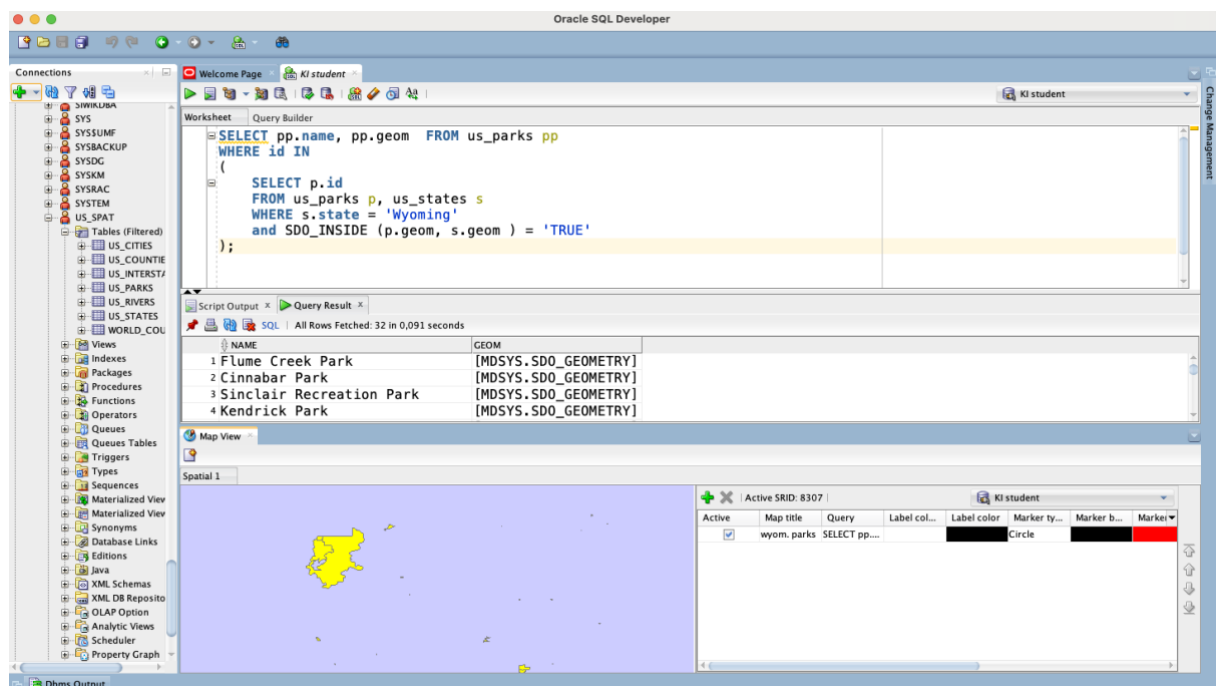
Znajdź wszystkie parki (us_parks) których obszary znajdują się wewnątrz stanu Wyoming

SDO_INSIDE

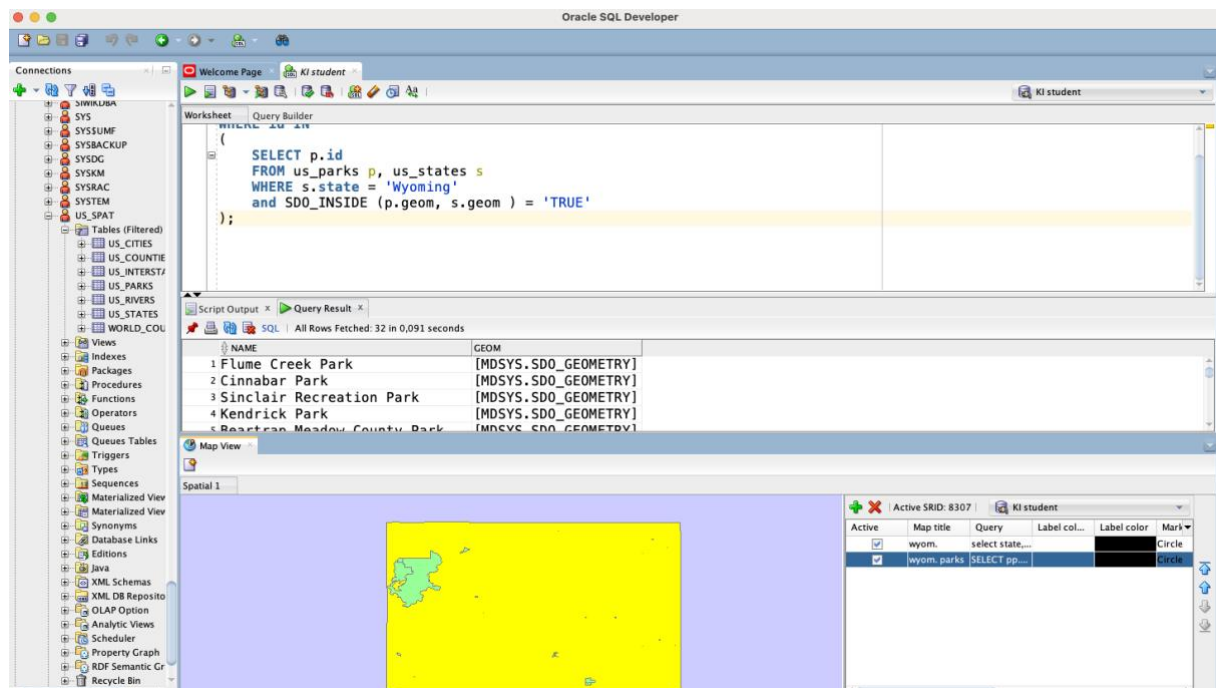
```
SELECT p.name, p.geom
FROM us_parks p, us_states s
WHERE s.state = 'Wyoming'
AND SDO_INSIDE (p.geom, s.geom ) = 'TRUE';
```

W celu wizualizacji na mapie użyj podzapytania

```
SELECT pp.name, pp.geom FROM us_parks pp
WHERE id IN
(
    SELECT p.id
    FROM us_parks p, us_states s
    WHERE s.state = 'Wyoming'
    and SDO_INSIDE (p.geom, s.geom ) = 'TRUE'
)
```



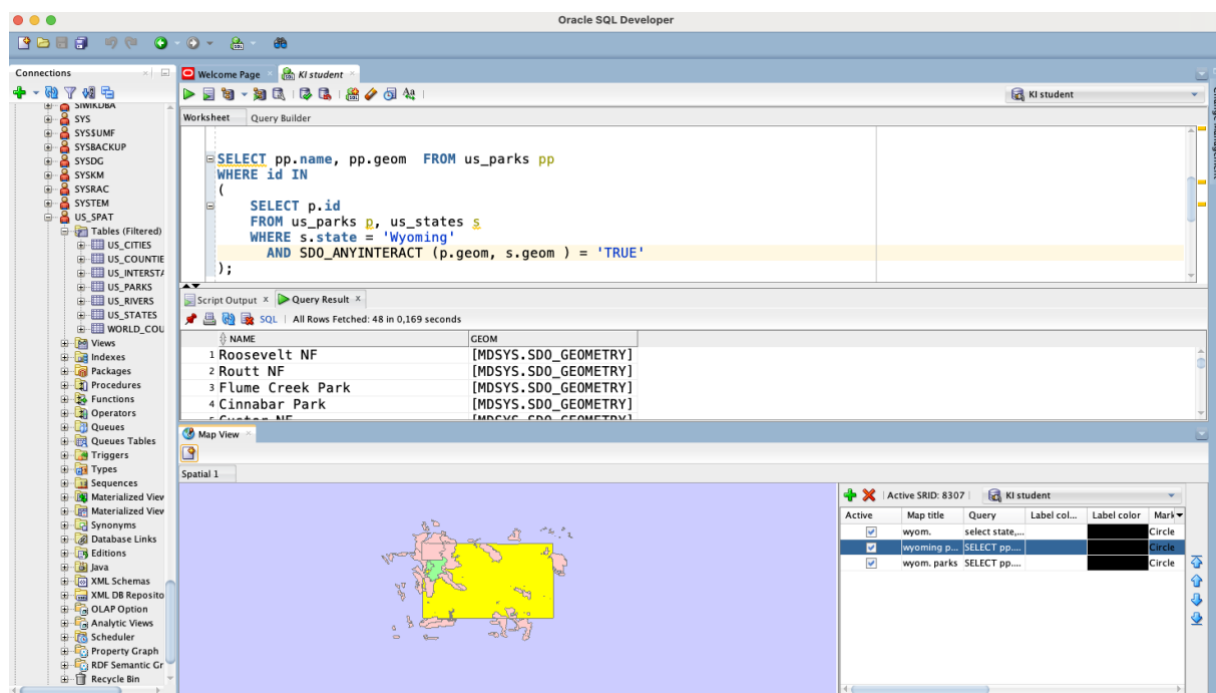
```
SELECT state, geom FROM us_states
WHERE state = 'Wyoming'
```



Porównaj wynik z:

```
SELECT p.name, p.geom
FROM us_parks p, us_states s
WHERE s.state = 'Wyoming'
AND SDO_ANYINTERACT (p.geom, s.geom) = 'TRUE';
```

W celu wizualizacji użyj podzapytania



Zadanie 4

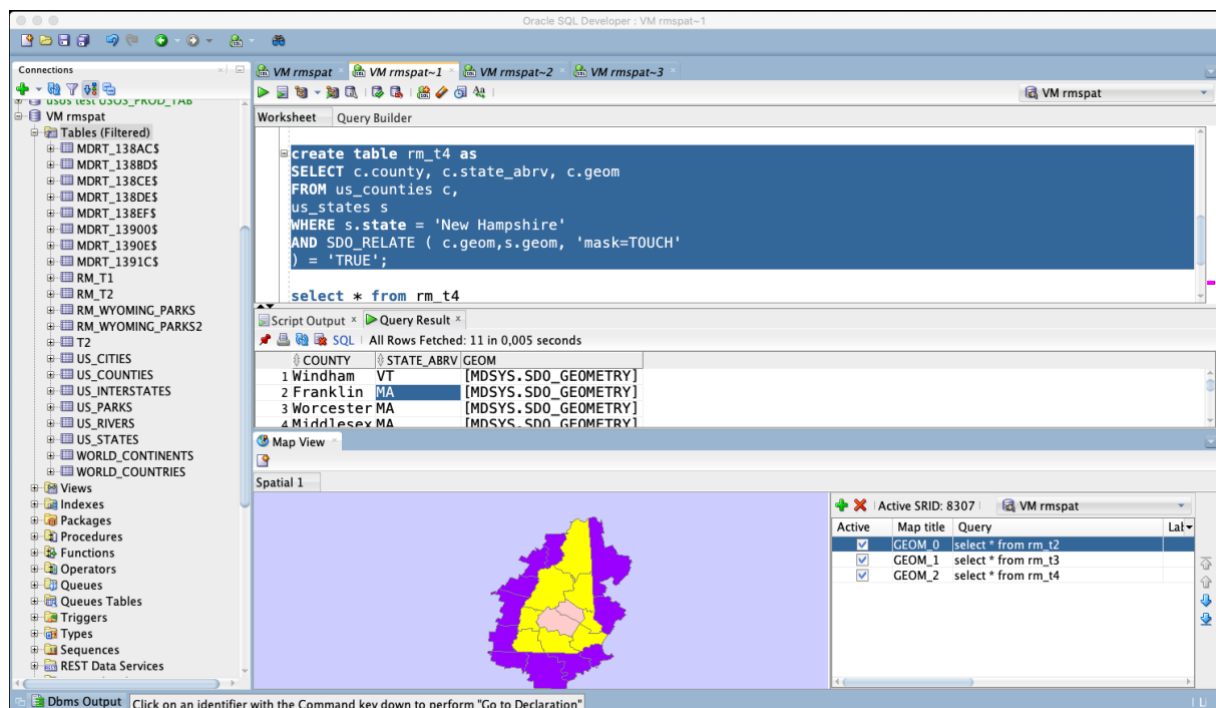
Znajdź wszystkie jednostki administracyjne (us_counties) wewnątrz stanu New Hampshire

```
SELECT c.county, c.state_abrv, c.geom
FROM us_counties c, us_states s
WHERE s.state = 'New Hampshire'
AND SDO_RELATE (c.geom,s.geom, 'mask=INSIDE+COVEREDBY') = 'TRUE';
```

```
SELECT c.county, c.state_abrv, c.geom
FROM us_counties c, us_states s
WHERE s.state = 'New Hampshire'
AND SDO_RELATE ( c.geom,s.geom, 'mask=INSIDE') = 'TRUE';
```

```
SELECT c.county, c.state_abrv, c.geom
FROM us_counties c, us_states s
WHERE s.state = 'New Hampshire'
AND SDO_RELATE ( c.geom,s.geom, 'mask=COVEREDBY') = 'TRUE';
```

W celu wizualizacji danych na mapie należy użyć podzapytania (podobnie jak w poprzednim zadaniu)



Zadanie 5

Znajdź wszystkie miasta w odległości 50 mili od drogi (us_interstates) I4

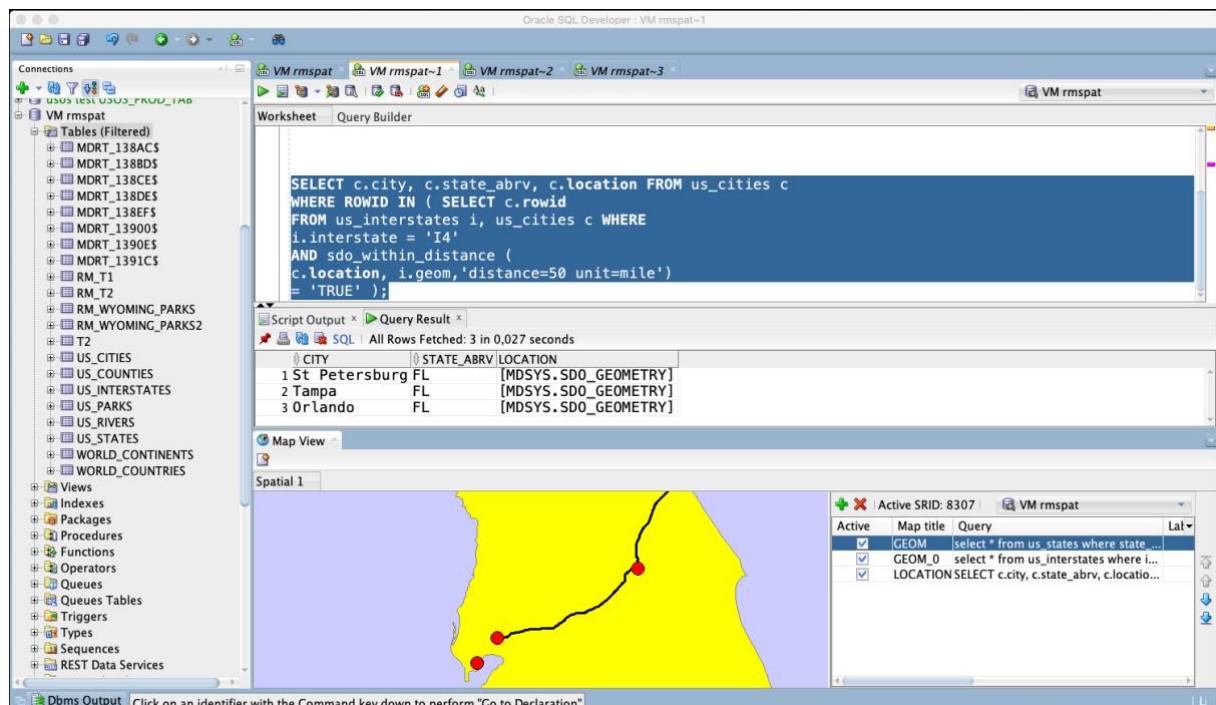
```
SELECT * FROM us_interstates
```



```
WHERE interstate = 'I4'
```

```
SELECT * FROM us_states  
WHERE state_abrv = 'FL'
```

```
SELECT c.city, c.state_abrv, c.location  
FROM us_cities c  
WHERE ROWID IN  
(  
  SELECT c.rowid  
  FROM us_interstates i, us_cities c  
  WHERE i.interstate = 'I4'  
  AND sdo_within_distance (c.location, i.geom, 'distance=50 unit=mile')  
  = 'TRUE'  
);
```



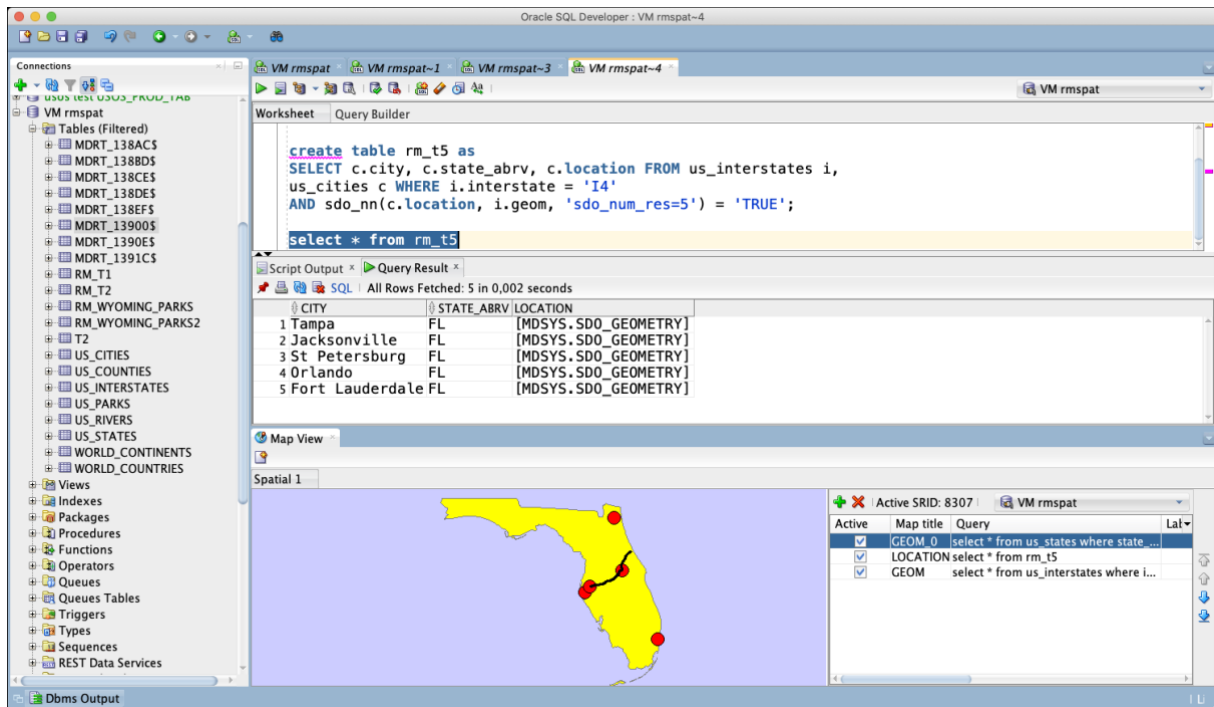
Dodatkowo:

- Znajdź wszystkie drogi które przecinają rzekę Mississippi
- Znajdź wszystkie miasta w odległości od 15 do 30 mil od drogi 'I275'
- ltp. (własne przykłady)

Zadanie 6

Znajdź 5 miast najbliższych drogi I4

```
SELECT c.city, c.state_abrv, c.location FROM us_interstates i,  
us_cities c WHERE i.interstate = 'I4'  
AND sdo_nn(c.location, i.geom, 'sdo_num_res=5') = 'TRUE';
```



Dodatkowo:

- Podaj 3 parki narodowe do których jest najbliżej z Nowego Jorku, oblicz odległości do tych parków
- Znajdź 5 najbliższych dużych miast (o populacji powyżej 300 tys) od drogi 'I170'
- Itp. (własne przykłady)
 - np. przetestuj działanie funkcji
 - sdo_intersection, sdo_union, sdo_difference
 - sdo_buffer
 - sdo_centroid, sdo_mbr, sdo_convexhull, sdo_simplify

Zadanie 7

Wykonaj kilka własnych przykładów/analiz